Gabrielle Johnson’s Week 7 Final Deliverable

Verdict

I am using the **class weighted Histogram Gradient Boosting (HGB) model** for the AI assisted triage system. It most directly addresses the primary safety objective; correctly identifying esi 2,3 and 4 patients well and minimising under triaging whilst generating predictions that are quick and explainable.

Dataset Summary

The dataset contains 55,121 emergency department patient encounters across 225 columns, approximately 200 of which are chief complaint columns. The remaining columns were grouped by clinical vitals, demographic features, administrative features, and the ESI levels. Potential leakage variables, such as disposition, were kept separate because they occur after triage and should not be used for prediction.

The dataset is adult only, with approximately 32% being over the age of 65. In terms of demographics, women are the majority of the patients recorded at 58%, additionally, 80% of patients are non-hispanic with 53% being of the White/Caucasian race.

Triage acuity was recorded using the Emergency Severity Index (ESI), which ranges from 1 to 5, where 1 represents the most urgent cases and 5 represents the least urgent cases. Nearly half (49%) of all encounters were classified as ESI 3, while fewer than 1% were classified as ESI 1.

The dataset also includes routinely collected triage vital signs, including heart rate, systolic blood pressure, diastolic blood pressure, respiratory rate, oxygen saturation, temperature, and glucose. Vital signs were reviewed using both plausible clinical limits and normal reference ranges. 29 patients had clinically implausible vital signs which were flagged as possible data quality issues.

The most common chief complaint was abdominal pain, representing 12.8% of encounters, followed by “other” at 7.8%. Although no demographic variables, vital signs, or ESI levels were missing, 52 patients had no recorded chief complaint. These cases were reviewed separately because the absence of a chief complaint may reflect documentation gaps rather than true absence of symptoms.

Metric Justifications

The primary evaluation metric for this project is per class recall across ESI levels 2–4, whilst also preventing under triaging especially for esi 2 as those patients may have subtle symptoms that can easily slip them into lower categories. For example an esi 2 patient cannot afford the same wait time as an esi 3 patient.

Although ESI 1 represents the highest clinical priority such cases are often clinically apparent at presentation. The model's greatest value lies in the more ambiguous middle categories, such as distinguishing an emergent ESI 2 patient from a stable ESI 3 patient or mid acuity esi 4 from a relatively stable esi 5. Recall is prioritised because it directly measures how many patients within each ESI category are correctly identified. Accuracy cannot capture this concern on an imbalanced dataset because 49% of encounters are ESI 3 and fewer than 1% are ESI 1, a model can score above 60% accuracy while failing to correctly prioritise a single critical patient. Therefore, reporting recall separately for each class prevents the ESI 3 category from dominating and masking poor performance on less common classes. Precision will still be reported alongside recall to ensure that improvements in identifying urgent patients do not result in excessive over triage/false alarms.

Benchmark Table

Model	Acc.	Prec.	Rec.	Macro-F1	Train (s)	Inf. (ms)	Explainable < 1 min? (method)
LogReg (base)	0.667	0.662	0.652	0.493	19.2	0.005	High — read coefficients
LogReg (+demo)	0.686	0.683	0.674	0.510	6.8	0.006	High — read coefficients
LogReg (+demo+FE)	0.690	0.687	0.678	0.520	15.1	0.003	High — read coefficients
LogReg weighted (+demo+FE)	0.668	0.662	0.682	0.481	16.7	0.003	High — read coefficients
Decision tree (base)	0.614	0.635	0.529	0.349	1.6	0.002	High — trace the tree path
Decision tree (+demo)	0.630	0.627	0.585	0.391	2.1	0.002	High — trace the tree path
Decision tree (+demo+FE)	0.627	0.624	0.583	0.394	2.3	0.001	High — trace the tree path
Neural network (MLP)	0.688	0.690	0.670	0.505	9.4	0.005	Low — SHAP only

Model	Acc.	Prec.	Rec.	Macro-F1	Train (s)	Inf. (ms)	Explainable < 1 min? (method)
Random forest	0.649	0.659	0.597	0.394	58.3	0.108	Medium — feature importances / SHAP
HGB	0.692	0.689	0.676	0.519	51.9	0.156	Low — SHAP only
HGB (weighted)	0.636	0.651	0.616	0.415	8.4	0.020	Low — SHAP only

Model	Accuracy	Macro-F1	ESI-2 recall	ESI-3 recall	ESI-4 recall	Under-tri. ESI-2	Under-tri. ESI-3	Under-tri. ESI-4
Weighted HGB	**0.636**	**0.415**	**0.797**	**0.620**	**0.430**	**19.6%**	**4.9%**	**2.9%**
Logistic regression + features	0.690	0.520	0.625	0.779	0.630	37.4%	8.3%	1.5%
Neural network (MLP)	0.688	0.505	0.604	0.797	0.609	39.5%	8.2%	1.8%
Decision tree	0.630	0.391	0.540	0.778	0.438	46.0%	7.5%	1.1%
Voting ensemble	0.681	0.440	0.758	0.693	0.573	24.0%	6.5%	2.2%
Stacking ensemble	0.675	0.484	0.786	0.651	0.600	21.3%	7.3%	1.7%

Three reasons for selecting the weighted HGB model

1. **It performs best where patient safety matters most**

   The model correctly identifies 79.7% of patients who are truly ESI 2, the highest rate among the models tested. It also has the lowest ESI 2 and 3 under triage rate at 19.6% and 4.8% respectively. It recognizes approximately four out of every five ESI 2 patients and is less likely than the other models to assign these urgent patients a lower priority category.

2. **It is fast and practical for real-time triage.**
   The model produces a prediction within seconds as it operates as one model rather than a chain of several models. Moreover, apart from being fast, it will also be easier to install, update, and monitor than an ensemble.
3. **Its predictions can be explained to clinicians.**
   The model can show which patient characteristics such as vital signs, age, arrival method, or chief complaint contributed most to a prediction using SHAP.

Three reasons against the weighted HGB model

1. **It performs poorly at identifying ESI 4 patients.**
   The model correctly identifies only 43% of patients who are truly ESI 4. This is considerably lower than logistic regression at 63% and the stacking ensemble at 60%. Although the weighting helps protect urgent ESI 2 patients, it reduces the model's ability to distinguish ESI 4 patients accurately.
2. **Excessive over triaging**
   The model assigns the correct ESI level in 63.6% of cases. Its emphasis on avoiding under-triage also produces more over-triage. Although this is generally safer than assigning an urgent patient a lower priority, it may increase unnecessary staff workload, hospital resources, and false alerts.
3. **It is not the easiest model to explain.**
   Logistic regression provides a relatively straightforward indication of how each patient characteristic influences the prediction. The weighted HGB model is more complex and requires an additional explanation method, such as SHAP, to show why it recommended a particular ESI level.

Risks

1. **Important information may be missing for under triaged ESI 2 patients.**
   The model assigns approximately one in five true ESI 2 patients to a less urgent category. These patients often resemble ESI 3 patients in the available data: their vital signs may be relatively normal, obvious warning signs may be absent, and their complaints may be vague. Therefore, an ESI 2 decision may depend also on medical history, comorbidities or previous presentations, information that is not fully captured in the current dataset. Future versions should therefore include richer clinical history and possibly information from the nurses' triage note/free text.
2. **Mid acuity Ambuguity**
   The middle ESI categories can be difficult to distinguish consistently. Hence, it is important to review difficult/flagged cases with more than one clinician
3. **The effect of over triaging**
   Reducing under triaging generally causes more patients to be assigned to higher acuity categories. Although this is usually safer than missing an urgent patient, it may increase staff workload, hospital resources, and unnecessary alerts. Therefore, the model should display its top reasons and let the nurse confirm or override for uncertain cases.